

UNIT V

VIRTUAL MACHINES AND MOBILE OS

Course Objective: To be Familiar with the Basics of Virtual Machines and Mobile OS.

Course Outcome: Compare iOS and Android Operating Systems.

Bloom's taxonomy level: k1, k2

PART - A

S.No.	Question	BTL
1	State virtual machine. Virtual machine abstracts the hardware of a single computer (the CPU, memory, disk drives, network interface cards, and so forth) into several different execution environments, thereby creating the illusion that each separate environment is running on its own private computer.	K1
2	Recall virtualization. Virtualization is the creation of a virtual rather than actual version of something. such as an operating system, a server, a storage device or network resources.	K1
3	Why virtualization is required? Virtualization reduces the number of physical servers, reducing the energy required to power and cool them. It saves time. It's also much faster to deploy a virtual machine than it is to deploy a new physical server. Reduce desktop management headaches.	K2
4	Name the various types of Virtualizations. • Full virtualization • Para virtualization • Hardware-assisted virtualization	K1
5	State Hypervisor. A hypervisor or virtual machine monitor (VMM) is computer software, firmware or hardware that creates and runs virtual machines. A computer on which a hypervisor runs one or more virtual machines is called a host machine, and each virtual machine is called a guest machine.	K1
6	List the two types of Hypervisors. There are two types of hypervisors: • Type 1 • Type 2 Type 1 hypervisors run directly on the system hardware. They are often referred to as a "native" or "bare metal" or "embedded" hypervisors in vendor literature. Type 2 hypervisors run on a host operating system.	K1
7	Recall para virtualization. Para virtualization is a computer hardware virtualization technique that allows virtual machines (VMs) to have an interface similar to that of the underlying or host hardware. With paravirtualization, the guest OS is modified, so it knows that it is running in a virtualized environment on top of a hypervisor (the hardware running the VM) and not on the physical hardware.	K1

8	Difference between Full Virtualization and Para virtualization		K2
	Full Virtualization	Para virtualization	
	Virtual machines permit the execution of the instructions with the running of unmodified OS in an entirely isolated way.	Virtual machine does not implement full isolation of OS but rather provides a different API which is utilized when OS is subjected to alteration.	
	Less secure.	More secure.	
	Uses binary translation and a direct approach as a technique for operations	Uses hypercalls at compile time for operations.	
	Slow than paravirtualization in operation.	Faster in operation than full virtualization	
	More portable and compatible.	Less portable and compatible.	
9	State XEN. Xen is a virtual machine monitor (VMM) for x86-compatible computers. Xen can securely execute multiple virtual machines, each running its own OS. on a single physical system with close-to-native performance, Xen is open source, is released under terms of the GNU General Public License.		K1
10	When does the virtual machine sprawl occurs? Virtual machine sprawl occurs when there are so many virtual machines on a system that their use, history, and state become confusing and difficult to track.		K2
11	Recall Hypercall. A hypercall is to asynchronous call what a hypervisor is to an OS.A call by a user process or the operating system for the hypervisor to perform some service required by the operating system of process.		K1
12	List the advantages of Virtual machine. - Multiple OS environments can exist simultaneously on the same machine, isolated from each other. - Virtual machine can offer an instruction set architecture that differs from real computers. - Easy maintenance, application provisioning, availability and convenient recovery.		K1
13	Mention the advantages of Virtualization. - Better utilization of computer hardware. - More securable and manageable than one kernel running many applications. - Reduce complexity of hardware. - Consumes less power and less space. - Requires less maintenance - Extended life for installed software		K1
14	State guest operating system. A guest OS is the software installed on either a virtual machine (VM) or partitioned disk that describes an operating system that is different than the host operating system.		K1
15	Why does user need Mobile OS? - A mobile operating system is an operating system that helps to run other application software on mobile devices. - A mobile OS is responsible for identifying and defining mobile device features and functions, including keypads, application synchronization, email, thumb wheel and text messaging.		K1

	A mobile OS is similar to a standard OS (like Windows, Linux, and Mac) but is relatively simple and light and primarily manages the wireless variations of local and broadband connections, mobile multimedia and various input methods.									
16	Distinguish between Type 1 and Type 2 Hypervisors. <table> <tr> <th>Type 1 Hypervisor</th> <th>Type 2 Hypervisor</th> </tr> <tr> <td>It is also known as Bare Metal or Embedded or Native Hypervisor.</td> <td>This is also known as Hosted Hypervisor.</td> </tr> <tr> <td>It works directly on the hardware of the host and can monitor operating systems that run above the hypervisor.</td> <td>The hypervisor is installed on an operating system and then supports other operating systems above it (runs on top of the supported OS).</td> </tr> <tr> <td>It's completely independent from the Operating System</td> <td>It is completely dependent on host Operating System for its operations.</td> </tr> </table>	Type 1 Hypervisor	Type 2 Hypervisor	It is also known as Bare Metal or Embedded or Native Hypervisor.	This is also known as Hosted Hypervisor.	It works directly on the hardware of the host and can monitor operating systems that run above the hypervisor.	The hypervisor is installed on an operating system and then supports other operating systems above it (runs on top of the supported OS).	It's completely independent from the Operating System	It is completely dependent on host Operating System for its operations.	K2
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17	Recall iOS. iOS is a Mobile Operating System that was developed by Apple Inc. for iPhones, iPads, and other Apple mobile devices. iOS is the second most popular and most used Mobile Operating System after Android. The structure of the iOS operating System is Layered based. Its communication doesn't occur directly. The layer's between the Application Layer and the Hardware layer will help for Communication. The lower level gives basic services on which all applications rely and the higher-level layers provide graphics and interface-related services.	K1								
18	List the different layers in iOS. - Core OS - Core Services - Media Layer - Cocoa Touch	K1								
19	Outline the features of iOS operating System. - Highly Securer than other operating systems. - iOS provides multitasking features like while working in one application we can switch to another application easily. - iOS's user interface includes multiple gestures like swipe, tap, pinch, Reverse pinch. - iBooks, iStore, iTunes, Game Center, and Email are user-friendly. - It provides Safari as a default Web Browser. - It has a powerful API and a Camera. - It has deep hardware and software integration	K1								
20	State Services Layer. Service layer is the middle layer between presentation and data store. It abstracts business logic and data access, The idea behind such a layer is to have an architecture which can support multiple presentation layers.	K1								
21	Recall Android OS. Android OS is a Linux-based mobile operating system that was developed by Google, primarily runs on smart phones and tablets. The Android platform includes an operating system based upon the Linux kernel, a GUI, a web browser and end-user applications that can be downloaded.	K1								
22	Name the features of android operating systems. - Near Field Communication (NFC) - Infrared Transmission - Automation - Wireless App Downloads - Storage and Battery Swap - Custom Home Screens	K1								

- Widgets
- Custom ROMs

PART-B

S.No.	Question	BTL *
1	Elaborate the benefits, features and building blocks of virtual machine. (16)	K1
2	Discuss the various virtual machine technologies. (16)	K1
3	Illustrate the methods and types used to implement virtualization with an example. (16)	K2
4	Elaborate the Mobile OS components with a neat diagram. (16)	K2
5	Interpret the different layers of iOS architecture with a neat diagram. (16)	K2
6	Interpret the different layers of android architecture with a neat diagram. (16)	K2
7	Elaborate SDK frameworks with a neat diagram. (16)	K2
8	Case Study: Virtual Machines and Mobile Operating Systems in a Software Company (16) A small software development company wants to improve its IT infrastructure. Earlier, the company used separate computers to test different operating systems. This increased hardware cost and maintenance. To solve this problem, the company decided to use virtual machines. By using virtualization software like Oracle VM VirtualBox, the company installed multiple operating systems such as Windows and Linux on a single physical computer. Each operating system runs in a separate virtual machine, allowing developers to test applications easily without needing multiple devices. This helped the company reduce cost and improve resource utilization. At the same time, the company started developing mobile applications for customers. The apps were designed to run on mobile operating systems like Android and iOS. Developers used the Android SDK and iOS SDK to build and test the applications. Through virtualization and mobile operating systems, the company was able to test software efficiently, support multiple platforms, and provide better mobile services to its users. Based on the above case study, explain the following: 1. History and benefits of Virtual Machines. 2. Building blocks and types of Virtual Machines. 3. Concept of virtualization. 4. Architecture of Android and iOS mobile operating systems. 5. Role of SDK frameworks in mobile application development.	K3